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Finance, Investment and Risk
Level: Intermediate • Rev. 2026

Time Value of Money, Discounting, and Compound Interest

Summary: Why a naira today is worth more than a naira tomorrow. Calculate Present Value, Future Value, Net Present Value (NPV), and internal hurdle rates.

Learning Objectives

- Understand the core principle of the Time Value of Money (TVM).
- Calculate Future Value (FV) with compound interest.
- Discount future commercial cash flows to calculate Present Value (PV) and Net Present Value (NPV).
- Adjust investment hurdles for high-inflation macroeconomic environments.

Core Principles & Key Takeaways

- A currency unit today is worth more than one in the future due to inflation and reinvestment return.
- Compound interest accelerates capital growth exponentially over long investment horizons.
- Discount future project cash flows using a rate that reflects inflation and commercial risk.

1. Core Principle of Time Value of Money

Money available at the present time is worth more than the identical sum in the future due to its potential earning capacity (interest and investment yields) and the eroding effect of inflation.

2. The Mathematical Formulas

$$\text{Future Value (FV)} = \text{PV} \times (1 + r)^n$$
$$\text{Present Value (PV)} = \frac{\text{FV}}{(1 + r)^n}$$

Where:
* **PV:** Present Value (starting principal).
* **FV:** Future Value after compounding.
* **r:** Annual interest rate or discount rate per period.
* **n:** Number of compounding periods (years).

$$\text{Net Present Value (NPV)} = \sum_{t=1}^{n} \frac{CF_t}{(1 + r)^t} - \text{Initial Investment}$$
If $\text{NPV} > 0$, the project generates value in excess of the alternative hurdle rate and should be undertaken.

Self-Assessment & Review Questions

Q1: If you invest ₦1,000,000 at a compound interest rate of 20% per year, what is the value after 2 years?

A. ₦1,400,000
B. ₦1,440,000

C. ₦1,200,000
D. ₦1,500,000

Correct Answer: B • FV = PV * (1 + r)^n = ₦1,000,000 * (1 + 0.20)^2 = ₦1,000,000 * 1.44 = ₦1,440,000.

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